Paleomagnetism of the Great Slave Supergroup, Northwest Territories, Canada: the Tochatwi Formation

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The Tochatwi Formation comprises some 800 m of fine-grained red to buff sandstone in the upper part of the Great Slave Supergroup. Recent geological work indicates an age of 1700 m.y., but this is not yet known with certainty. Results from 29 sampling sites stratigraphically covering the Tochatwi Formation are presented. Standard paleomagnetic techniques indicate that post-folding remagnetization is common, and this is confirmed by analysis of the magnetic vectors removed as partial thermal demagnetization proceeds. Two phases of remagnetization are recognized, one of which can be attributed to nearby Mackenzie-age intrusions. The other phase is equivalent to a remagnetization observed by other workers in Kahochella Group strata at localities 70 km away. As yet, the source of this remagnetization event cannot be identified. Eight sampling sites have escaped total remagnetization and an earlier, pre-folding remanence has been isolated from these ($D = 030, I = -11, k = 14, \alpha_{95} = 15°$). The corresponding pole position (144W, 18S) is remote from those deduced from North American rocks of similar age, and possible explanations of this problem are discussed.

La formation de Tochatwi est constituée de 800 m d'un grès à grain fin de couleur rouge à chamois dans la partie supérieure du supergroupe Great Slave. Des travaux géologiques récents donnent un âge de 1700 Ma bien que ce ne soit pas encore définitif. Les résultats de 29 échantillons couvrant stratigraphiquement la formation de Tochatwi sont indiqués. Les techniques paléomagnétiques standard indiquent qu'après déformation une remagnétisation est fréquente, ceci étant confirmé par l'analyse des vecteurs magnétiques éliminés à mesure que la démagnétisation thermale partielle progresse. On reconnaît deux phases de remagnétisation dont l'une peut être attribuée aux intrusions d'âge Mackenzie qui sont tout près. L'autre phase équivaut à une remagnétisation reconnue par d'autres chercheurs dans les strates du groupe de Kahochella à une distance d'environ 70 km. A date, on ne peut identifier la source de cet événement de remagnétisation. Huit endroits échantillonnés ont échappé à la remagnétisation totale, une rémanence antérieure à la déformation ayant été isolée dans ces échantillons ($D = 030$, $I = -11$, $k = 14$, $\alpha_{95} = 15°$). La position correspondante du pôle (44° W, 18° S) est loin des positions déduites de roches d'Amérique du Nord d'âge semblable et on discute d'explications plausibles à ce problème.

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Introduction

The Great Slave Supergroup comprises some 12 000 m of unmetamorphosed sedimentary and volcanic rocks preserved in the East Arm Fold Belt of Great Slave Lake (Hoffman 1968). As many of the sedimentary strata contain red beds the area is well suited to studying the Proterozoic paleomagnetic record.

The Christie Bay Group is the uppermost of four groups (Sosan, Kahochella, Pethei, Christie Bay) into which Hoffman (1969) divided the Great Slave Supergroup. It consists of four conformable formations, which are, in order of decreasing age, the Stark, Tochatwi, Portage Inlet, and Pearson Formations. Paleomagnetic results from the oldest of these, the Stark Formation, are reported in a companion paper (Bingham and Evans 1976). The present paper concerns the Tochatwi Formation, which conformably overlies the Stark Formation. Geological details can be found in a comprehensive report by Hoffman (1968). Briefly, the Tochatwi Formation consists of fine-grained red-to-buff sandstone up to 800 m thick. Cross-bedding, mud cracks, and ripple marks are common and a few beds of shale or conglomerate occur. Hoffman (1969) concluded that the formation is largely fluviatile, except at the base, which may represent a tidal marine environment.

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In many localities, strata of the Great Slave Supergroup are tightly folded. The sedimentary rocks comprising the Et-then Group rest uncomfortably on the truncated folds of the Great Slave Supergroup and are themselves only gently folded (Barnes 1951). This implies that the bulk of the folding of the Christie Bay Group and older strata pre-dates the deposition of Et-then Group rocks, an observation that will become important in connection with the ages of the various magnetizations found in these rocks.

Methods and Sampling Details

Standard orientation, measurement, and demagnetization procedures were employed (see Bingham and Evans 1976, for details). The study involved about 1000 measurements of remanence.

Two to six samples were collected at each of 29 stratigraphically distinct sites (Fig. 1). Where possible, these samples were drilled, but at four sites the friable condition of the material precluded drilling, and hand samples were therefore collected. In sedimentary strata sampling was carefully confined to a single stratigraphic horizon at each sampling site. The sites are located in four separate areas, representing four stratigraphic sections: section A (6 sites), section F (5 sites), section G (2 sites), and section N (16 sites). The N section lies about 14 km east of the other sections (Fig. 1). It spans the entire Tochatwi Formation, which at this locality is approximately 600 m thick. The A and G sections each contain one site in igneous rocks probably intruded during the Mackenzie igneous events at about 1200 m.y. (Fahrig and Jones 1969); site AA is located in a 1.5 m thick shallowly dipping intrusive sheet cutting Tochatwi rocks; site AB is a baked contact less than 0.5 m from this intrusion; sites AD, AC, AE, and AF lie roughly 25, 75, 125, and more than 200 m away from the contact. As these distances were measured horizontally across the outcrop and as the sheet dips at about 20° their perpendicular separation from the intrusion is difficult to determine, but is likely to be considerably less than the figures given. In fact, if the sheet was originally irregular in the area from which it is now eroded the separation is unknown. A total stratigraphic thickness of about 100 m is represented, but how it correlates with sections N and F is not known. Site GA is in a large ($\sim$ 50 m thick) sill, whereas GB is in Tochatwi rocks about 50 m distant; the contact itself is obscured by soil and vegetation. The F section represents the lowest 90 m of Tochatwi sediments, and is thus laterally equivalent to the lower part of section N.

From each site two pilot specimens were selected: one for detailed alternating field (a.f.) and one for detailed thermal demagnetization experiments. In all cases, the thermal treatment is preferred because where a.f. and thermal results disagree significantly, the thermal result is always farther from the natural remanent magnetization (NRM) direction. In fact, the a.f. result often differs little from the NRM. It appears that thermal treatment more successfully removes secondary components of magnetization.

According to the quality of the thermal end-points of pilot specimens one or several temperatures were chosen for blanket treatment of one specimen from every sample collected. For sites at which several temperatures were used the data obtained from that treatment yielding highest within-site precision were selected as the site mean (Irving *et al.* 1961).

Discussion

Directions of Magnetization

For discussion, it is convenient to divide the results according to sampling locality or section.

N Section

During pilot thermal demagnetization 16 specimens (one from each site) move systematically along great circles by 20 to 50°, or more, before reaching reasonable end-points (McElhinny and Gough 1963) between 500 and 600 °C. They also show markedly distributed blocking temperature spectra (Irving and Opdyke 1965), with intensities after 200 °C heating lying between 5 and 35% of NRM (Fig. 2). The vectors removed by the first heating step (200 °C) are loosely grouped (266, +85, $N = 16$, $k = 8$, $\alpha_{95} = 14°$) about the present field at the sampling locality, suggesting that a recently acquired magnetization is being removed.

On the basis of the pilot study, one specimen from each sample ($N = 46$) was heated to 500 and 600 °C. In four cases site means were random (Watson 1956*a*) at both temperatures; 11 of the remaining 12 possess higher within-site precision at 500 °C and these yield the data given in

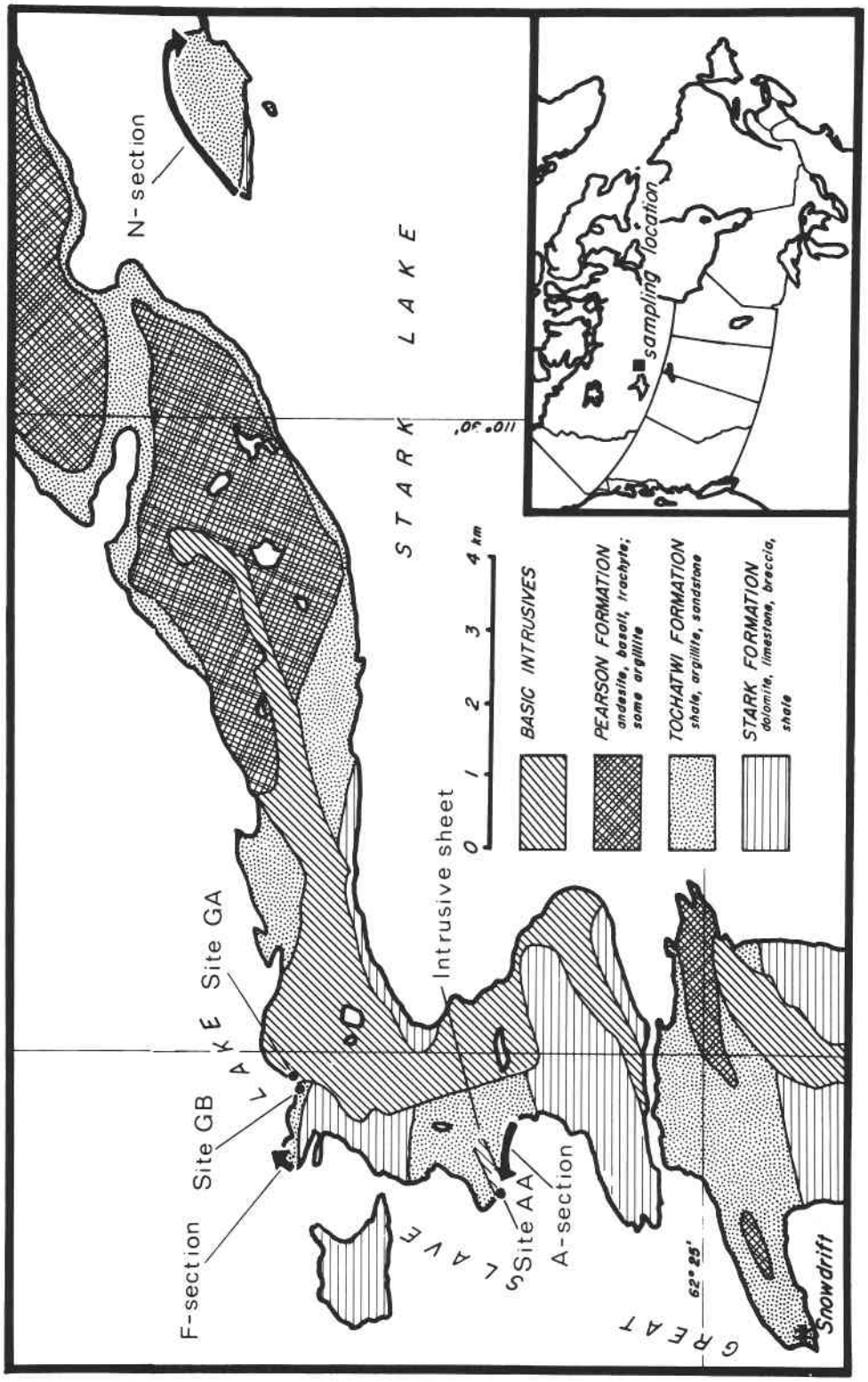


FIG. 1. Location map showing sampling locations. Heavy arrows point to stratigraphic tops of sections. Geology after Barnes (1951).

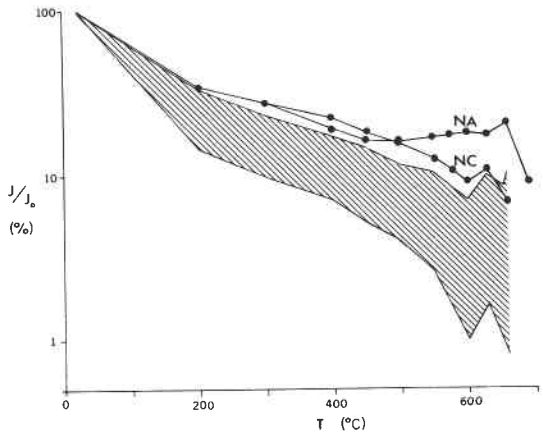


FIG. 2. Thermal demagnetization curves for N-section specimens. Sixteen specimens are represented, with the decay curves for 14 of them lying within the shaded zone.

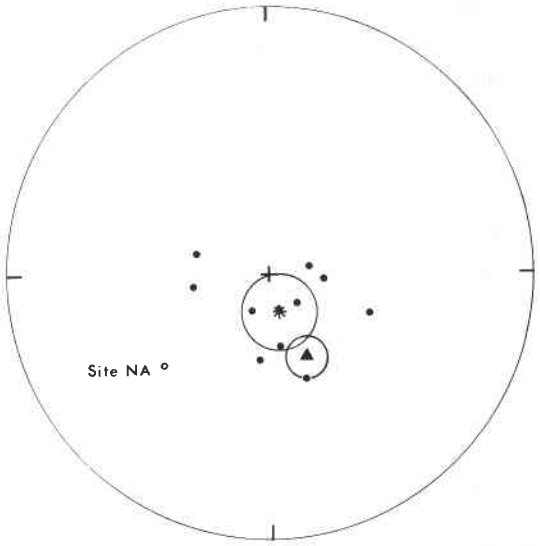


FIG. 3. Non-random site mean directions for the N-section before correction for geological dip. Solid (open) symbols on lower (upper) hemisphere. The mean of the 11 lower hemisphere directions is represented by the star. The triangle represents a remagnetization direction present in Kahochella Group strata (McMurry *et al.* 1973) (see text). Circles of 95% confidence are shown.

Table 1. The remaining site (NA) is distinct in that it required heating to 600 °C. This site also differs from the others in its blocking temperature spectrum, retaining 18% of its NRM at 600 °C, whereas the others range from 1 to 9% (Fig. 2). Figure 3 shows the 12 non-random site means *before* correction for local geological dip. Again, it will be noted that site NA is quite distinct. The other 11 sites form a reasonable group with a mean direction of 165, +78 (k = 16, $\alpha_{95} = 12°$), but after structural correction the scatter is much increased, and k decreases to 4.0. This decrease is significant at the 99% level (McElhinny 1964), indicating a negative fold test (Graham 1949).

The mean direction before unfolding lies only 16° from that reported by McMurry *et al.* (1973) and Reid (1972), for a post-folding remagnetization event revealed by detailed thermal demagnetization studies of the Kahochella Group. As the α_{95} circles overlap it seems possible that the two sets of data represent the same remagnetization event.

To test this idea further the vectors removed between 500 and 600 °C have been analysed. At three sites these were statistically random, but the remaining 13, including site NA, form a group with a mean direction of 142, +71 ($k = 24$, $\alpha_{95} = 9°$). Again, after structural correction scatter increases markedly with k decreasing to 4.8, leading to a negative fold test at the 99% level. Before structural correction the mean direction lies only 10° from that resulting

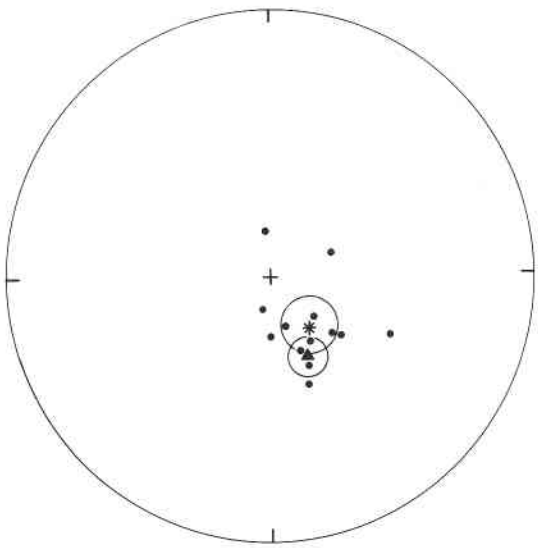


FIG. 4. Non-random site means for the vector removed from N-section samples between 500 and 600 °C, before correction for local geological dip. Symbols as in Fig. 3.

from the remagnetization event identified in Kahochella Group rocks and the circles of 95% confidence overlap (Fig. 4). The standard statistical test (Watson 1956*b*) indicates that we cannot assert that the two mean directions

TABLE 1. Site mean paleomagnetic directions for Tochatwi Formation

	Str	Dip	N	T	Before dip correction D	Before dip correction I	After dip correction D	After dip correction I	k	R
N-Section										
NP	013	43	2	500	160	56	134	24	—	1.9911
NO	004	37	3	500		*		*	4.3	2.5347
NN	001	42	3	500	261	66	103	71	8.8	2.7729
NM	345	27	3	500	287	66	012	76	54.1	2.9630
NL	300	22	3	500		*		*	4.6	2.5655
NK	299	23	3	500	187	64	128	81	111.6	2.9821
NJ	285	29	3	500		*		*	4.6	2.5672
NI	233	28	3	500		*		*	5.1	2.6044
NH	235	38	3	500	111	57	027	70	14.1	2.8586
NG	241	52	3	500	137	78	336	50	9.2	2.7815
NF	241	58	3	500	094	73	350	40	7.2	2.7212
NE	247	64	2	500	209	77	325	33	—	1.9879
ND	246	70	3	500	172	67	328	42	62.4	2.9680
NC	242	77	3	500	165	77	329	26	71.0	2.9718
NB	247	86	3	500	079	77	350	07	24.3	2.9177
NA	245	82	3	600	228	−47	196	06	76.5	2.9739
A-Section										
AA		—	4	500	245	08	—	—	517.1	3.9942
AB	089	30	5	500	244	05	244	−08	311.9	4.9872
AD	089	30	3	660	038	−65	019	−38	30.2	2.9338
AC	089	30	2	575	043	−59	025	−34	—	1.9964
AE	084	31	3	450	245	20	238	08	287.3	2.9930
AF	083	23	3	400	252	21	244	16	69.0	2.9710
G-Section										
GA		—	4	550	272	52	—	—	43.2	3.9306
GB	302	72	4	500	249	17	306	55	271.6	3.9890
F-Section										
FE	302	126	4	500	209	60	033	−06	44.1	3.9320
FD	300	133	4	450	210	52	030	−05	185.8	3.9839
FC	305	125	4	500	199	56	044	−02	67.7	3.9557
FB	298	128	4	300	217	53	022	−01	161.3	3.9814
FA	300	123	4	660	344	−48	239	01	36.8	3.9185

Str = strike measured clockwise from North.
Dip = dip of beds, measured clockwise when looking along strike. Dips > 90 indicate overturning.
N = number of samples (1 specimen each).
T = thermal cleaning temperature (°C).
D, I = declination and inclination.
$k = N-1/(N-R)$
R = resultant of N unit vectors.
* = site means random (Watson 1956*a*).

differ. Thus it would appear that this stratigraphic section records the legacy of the same post-folding remagnetization event as that found in the Kahochella Group sampled some 70 km to the east. At the present time not much can be inferred as to the nature of this event. Its areal spread favours a thermal process such as might accompany burial and uplift or widespread intrusive activity. In the latter category there exist two likely possibilities: Mackenzie-age intrusions, and the earlier quartz–diorite laccoliths, both of which occur commonly throughout the area. A Mackenzie reheating can be ruled out because the direction diverges by almost 70° from that necessary (Fahrig and Jones 1969; Irving *et al.* 1972). To date no paleomagnetic information is available on the quartz–diorite laccoliths, but an investigation is currently in progress. It will be of some interest to see if these intrusions are indeed the remagnetization source, as they are thought to have been emplaced during upper Christie Bay Group times (Hoffman 1969); in fact, they are not reported actually cutting any strata higher than the Stark Formation, which underlies the Tochatwi Formation. Thus they offer a potential means of considerably re-

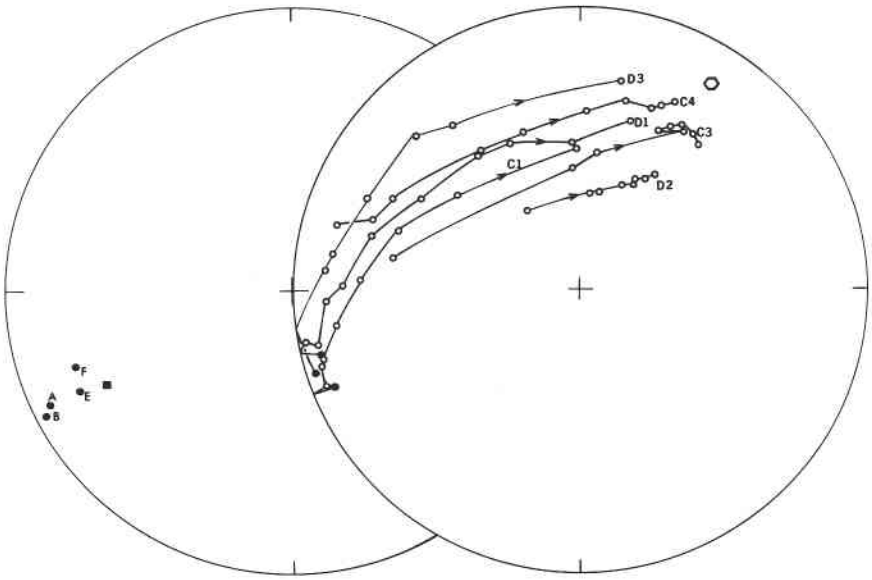


FIG. 5. (*a*, left) Site mean directions for A-section sites, before geological dip correction. The leading A's are omitted for clarity. The square represents the expected Mackenzie igneous events direction (Fahrig and Jones 1969) at this locality.

(*b*, right) Behaviour of directions of magnetization during progressive thermal treatment for samples from sites AC and AD, after correction for geological dip. The hexagon (upper hemisphere) represents the Stark Formation mean direction of magnetization (Bingham and Evans 1976). Symbols as in Fig. 3. Temperature steps shown are as follows: C1 and C3—0, 400, 450, 500, 550, 575, 600, 630, 660; C4—0, 200, 300, 400, 450, 500, 575, 600, 630; D1—0, 200, 300, 400, 450, 500, 550, 575, 600, 630, 660; D2—0, 400, 450, 550, 575, 600, 630, 660; D3—0, 500, 550, 575, 600, 630, 660. Erratic high temperature directions are omitted for clarity.

stricting the ages of the remagnetization itself and, in turn, an earlier magnetic component discussed below. A third possible source of reheating could be associated with the extrusion of the Pearson Volcanics, which are the youngest formation of the Christie Bay Group. Here again no data are available, but their restricted areal distribution renders them a less likely source for widespread reheating.

A Section

Site AA represents an intrusive sheet cutting Tochatwi strata. Pilot demagnetization indicated highly stable remanence, and a site mean direction southwesterly with shallow positive dip is found (Fig. 5*a*). Similar directions were obtained from sites AB, AE, and AF, and again angular movements were very small, except for two specimens, which at high temperatures begin to show systematic movement towards the northeast quadrant. The meaning of this behaviour becomes clear when the two remaining sites (AC, AD) are scrutinized. Two specimens from site AC migrate along great circles and reach end-points in the northeast quadrant, whereas a third specimen moves in like manner but fails to reach a good end point before its highest blocking temperatures are exceeded (Fig. 5*b*). From site AD, one specimen attains a similarly directed end-point and the other two migrate to the same direction at 660 °C. Thus sites AC and AD provide strong evidence for the presence of two magnetic components. The one with lower blocking temperatures apparently represents a remanence acquired at the time of heating by the intrusive sheet (or unexposed equivalents), whereas the high temperature component presumably reflects an earlier magnetization. On the other hand, sites AB, AE, and AF have been almost totally remagnetized.

The age of the intrusive sheet (site AA) can be reasonably well inferred from the magnetic data itself, as its direction of remanence lies close to that expected for Mackenzie age intrusions ($\sim$ 1200 m.y.; Fahrig and Jones 1969; Irving *et al.* 1972). This is further substantiated

by the site means for the baked sites (AB, AE, AF). Furthermore if the vectors removed between NRM and 500 °C are calculated this Mackenzie remagnetization can also be recognized in the partially reheated sites (AC, AD). The five site means for the removed vectors have a mean of 249, +29 (k = 35, α_{95} = 13°), which does not differ significantly from the expected Mackenzie direction (Fig. 6). Since the Mackenzie intrusions were emplaced after the folding the above comments obviously refer to directions of magnetization prior to any correction for local geological tilt.

The meaning and significance of the earlier magnetic component revealed at sites AC and AD are more conveniently discussed after the F and G section results are described.

G Section

During thermal demagnetization specimens from sites GA and GB show systematic movement and reach good end-points at or prior to 500 °C. Inspection of Table 1 indicates that the sedimentary site (GB) strongly suggests a Mackenzie reheating, but the intrusion itself is somewhat divergent (~ 30°), although not beyond the limits of reasonable secular variation.

F Section

Of six samples collected at site FA, two are excluded owing to inconsistent behaviour during test demagnetization runs (one exhibited erratic behaviour, and the other yielded an end-point diverging widely from the rest of the data). The remaining four show excellent end-points between 600 and 630 °C after having migrated along great circle paths by more than 90°. The other four sites (FB–FE) possess very good within—and between—site grouping at temperatures from 300 to 500 °C (Table 1). It will be noted that site FA has a magnetic polarity opposite to the other four sites. Inverting it and averaging all five yield a mean direction of 038, −03 (k = 31, α_{95} = 14°). This result represents the data after correction for local geological dip. Before dip correction (D = 199, I = +56, k = 36, α_{95} = 13°) they offer no evidence of remagnetization by either of the two possible causes identified above. The presence of a reversal emphasizes this point because both potential remagnetization sources possess only one polarity (Fahrig and Jones 1969; Reid 1972).

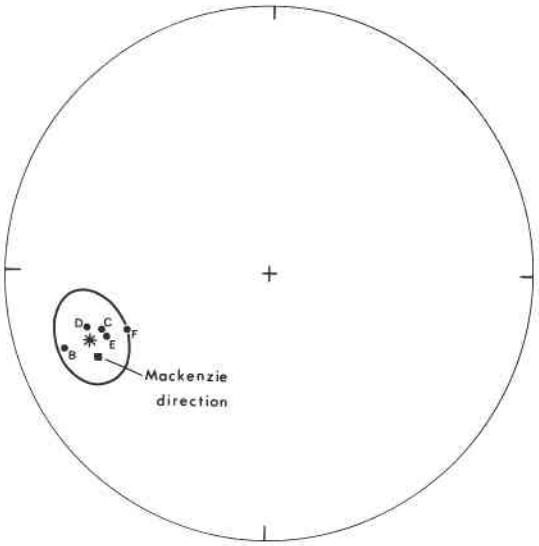


FIG. 6. A-section site means for the magnetic vectors removed between 0 and 500 °C (0–550° for D) showing the effect of Mackenzie-age reheating. Directions shown refer to the situation before correction for geological dip. Symbols as in Fig. 3.

Pole Position and Magnetic Polarity

Detailed thermal demagnetization studies coupled with analysis of the magnetic vectors removed by such treatment indicates that most of the Tochatwi Formation samples collected have undergone some form of remagnetization. However, eight sites appear to have escaped such influence. These are sites FA–FE, AC, AD, and NA, and their mean directions of remanence are illustrated in Fig. 7. Before dip correction they are widely scattered, and their resultant indicates randomness (Watson 1956*a*), but after allowing for local geological tilt a coherent group emerges with a mean direction of 030, −11 (k = 14, α_{95} = 15°). The between-site precision parameter increases by a factor of nine when bedding attitude is allowed for, thus providing strong evidence that magnetization preceded folding, which is known to have occurred before deposition of Et-then group strata. As the Stark direction is the same as the Tochatwi direction (Fig. 7), this in turn implies that the Stark magnetization is also pre-Et-then in age.

On the usual assumption of a geocentric dipole the mean direction maps to a paleomagnetic pole at 144° W, 18° S. However, it is stressed that Bingham and Evans (1976) have argued that a small crustal block containing

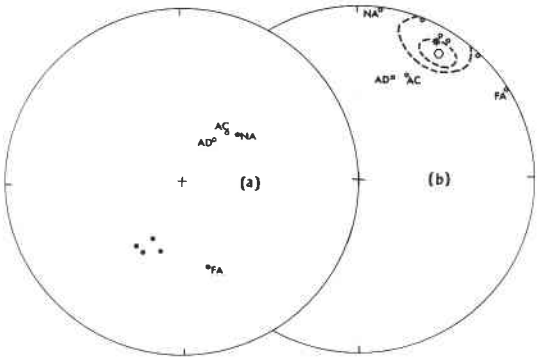


FIG. 7. Mean directions of magnetization for the eight Tochatwi sites retaining evidence of a pre-folding magnetization. Directions for sites NA and FA are inverted because of their polarity (see Table 1). (a) Before correction for geological dip. (b) After correction for geological dip. The star represents the mean direction for the 8 sites, and the hexagon represents the Stark Formation direction (Bingham and Evans 1976)—both on the upper hemisphere. Circles of 95% confidence are also shown. Other symbols as in Fig. 3.

the sampling sites of both Stark and Tochatwi Formations may have been rotated about a vertical axis. This implies that the pole positions obtained should not be incorporated into polar wander paths until verified by results from tectonically 'normal' areas.

Poor effective stratigraphic coverage resulting from widespread remagnetization means that very little can be deduced about geomagnetic polarities in Tochatwi times. One interesting point does emerge, however. Reference to figs. 2 and 8 in the companion paper (Bingham and Evans 1976) shows that the polarity of sites SEA and SEB, which occur at the very top of the Stark Formation, is normal (*i.e.* southwest and shallow). The same polarity is recorded at sites FA and NA, which are separated laterally by some 14 km but which both lie in the bottom 20–40 m of the Tochatwi Formation. Between sites FA and FB, a reversal occurs indicating that the normal polarity magnetozone which starts near the top of the Stark Formation occupies a limited stratigraphic thickness and was therefore probably short-lived.

Acknowledgments

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